

Metabolism & Eating Disorders

What is Metabolism?

Metabolism is how your body turns food and drink into energy. Our body needs food to fuel our cells and keep us alive and functioning. Through the process of metabolism, the body breaks down the food you eat into building blocks called fatty acids, amino acids and glucose. These building blocks are used to provide energy for your body to function.



What do we use energy for?

Your body requires energy for four main functions:

1. For your organs (such as lungs, heart, brain) to function. For example, your body generates heat to keep you warm, uses lungs to breathe, and keeps your heart beating to pump blood to the other organs. In fact, your brain alone uses about one third of your daily supply of building blocks, especially glucose. This component of metabolism is also known as *basal metabolic rate* and accounts for 60-80% of your total energy needs.
2. For growth and repair of body tissues when we are hurt or sick, including fighting infection and when the body is under stress.
3. To move your body during physical activity, including walking, cleaning, shopping, or playing sports.
4. To digest and break down food into its building blocks, also called the *thermic effect* of food.

What Affects Metabolic Rate?

The amount of food that we need depends on factors that increase or decrease your metabolic rate. These include:

- **Body size and the amount of muscle mass:** A larger body will require more energy, and muscle in particular requires a lot of energy to be maintained. A reduction in body or muscle mass will result in a reduced metabolic rate.
- **Dieting:** Dieting, fasting, and food restriction trigger your body to conserve energy for another time. This is a survival mechanism of your body, and is sometimes called the '*famine response*' (see our handout on [Starvation Syndrome](#)). Your body is very good at noticing an energy shortage and will reduce your metabolic rate to save energy.

- **Weight loss:** Any weight loss, especially when it is rapid, reduces metabolic rate.
- **Regularity of eating:** Eating regularly increases your metabolic rate whereas having long periods without food will decrease it. This is due to the thermic effect of food. It is helpful to eat at regular intervals; roughly every 3-4 hours (see our handout [Regular Eating for Recovery](#))
- **Genes:** Genes influence your resting metabolic rate, although this is not something you can change.
- **Sex:** In general, males have more muscle tissue than women. This means men may have a higher metabolic rate and require more energy.
- **Movement:** This includes all structured and planned activities as well as things you might not think about in day-to-day activity, such as cleaning or gardening.
- **Life stage:** Metabolic rate is higher during significant periods of growth and development such as puberty and pregnancy.
- **Infection and illness:** When you are unwell, your body has a higher metabolic rate and energy needs in order to fight the infection or illness.

How does eating disorder recovery affect metabolism?

When adequate nutrition returns, the body needs extra energy to heal and repair. Metabolism speeds up, which is sometimes called "**hypermetabolism**".

It is surprising how much energy is needed to repair muscles, tissues and internal organs. Early in recovery, a large portion of calories is directed to repairing internal systems, rather than weight restoration. This means you may need to eat much more than expected or more than others of a similar size for a period of time. Hypermetabolism is temporary, and as the body repairs itself and stabilises, energy needs gradually return to a more typical level.

Please remember that everyone's body responds differently and experiences may vary greatly. For information specific to your circumstances, we recommend consulting with a dietitian.

